Anubrata Das, Ph.D.

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Research Interests

My research spans Responsible AI and Human-AI collaboration. I develop (1) methods for Large Language Models (LLMs) to be fair and interpretable, and (2) post-training approaches that enable reasoning models to be effective collaborators for end users. I also collaborate with knowledge workers to understand their AI needs and translate them into insights for model development.

Employment

Autodesk Research — **New York, NY**
Principal Research Scientist — *03/2026 –*

- Project: Detecting and Mitigating Memorization in Geometry Generation Models

University of Texas at Austin — **Austin, Texas**
Postdoctoral Scholar, McCombs School of Business — *04/2025 – 02/2026*

- Mentors: Maytal Saar-Tsechansky, Liu Leqi
- Project: Building Trustworthy Reasoning Models to aid High-Stakes Decision Makers

Education

University of Texas at Austin — **Austin, Texas**
Ph.D., School of Information — *04/2025*

- Dissertation: Towards Human-Centered and Trustworthy Natural Language Processing
- Co-advisors: Matt Lease (iSchool, Computer Science); Junyi Jessy Li (Linguistics)

Indian Institute of Engineering Science and Technology Shibpur — **Kolkata, India**
Bachelor of Engineering, Department of Computer Science and Technology — *06/2015*

Awards and Honors

- **Rising Star in Data Science** — 2025
  - Organized by Stanford University,
    University of California, San Diego, and the University of Chicago
- **Best Paper Honorable Mention Award (CSCW 2024) (top 4%)** — 2024
- **Diversity & Inclusion Best Student Paper Award** — 2019
  - Awarded by the School of Information, University of Texas at Austin
- Spot Award - Mu Sigma Inc. — 2016
  - For developing an interactive visualization for stock market as a causal network
- Class of 1990 Excellence in Student Leadership Award — 2014
  - Awarded by the Global Alumni Association of BESU (now IIEST)

Publications [Google Scholar]

* denotes equal contribution

Conference and Journal Publications

13. On Localizing and Deleting Toxic Memories in Large Language Models
    Anubrata Das, Manoj Kumar, Ninareh Mehrabi, Anil Ramakrishna, Anna Rumshisky, Kai-Wei Chang, Aram Galstyan, Morteza Ziyadi, Rahul Gupta
    NAACL Findings, 2025
12. Finding Pareto trade-offs in fair and accurate detection of toxic speech
    Soumyajit Gupta, Venelin Kovatchev, **Anubrata Das**, Maria De-Arteaga, Matthew Lease
    iConference, 2025
11. Human-centered NLP Fact-checking: Co-Designing with Fact-checkers using Matchmaking for AI
    Anubrata Das*, Houjiang Liu*, Alexander Boltz*, Didi Zhou, Daisy Pinaroc, Matthew Lease, Min Kyung Lee
    CSCW, 2024
    Best Paper Honorable Mention (Top 4%)
10. The State of Human-centered NLP Technology for Fact-checking
    Anubrata Das, Houjiang Liu, Venelin Kovatchev, Matthew Lease
    Information Processing and Management (IPM Journal), 2023 (Impact Factor: 6.9)
9. True or false? Cognitive load when reading COVID-19 news headlines: an eye-tracking study
    Li Shi, Nilavra Bhattacharya, **Anubrata Das**, Jacek Gwizdka
    ACM SIGIR CHIIR, 2023
8. ProtoTEx: Explaining Model Decisions with Prototype Tensors
    Anubrata Das*, Chitrank Gupta*, Venelin Kovatchev, Matthew Lease, Junyi Jessy Li
    ACL Main, 2022
    (Acceptance rate: 701 of 3378 submissions, 20.8%)
7. The Need for Human-centered Design in Fact-checking Research
    Anubrata Das, Houjiang Liu, Venelin Kovatchev, Matthew Lease
    IPM Conference, 2022
6. Fairness in Information Access Systems
    Michael D Ekstrand, **Anubrata Das**, Robin Burke, Fernando Diaz
    Foundations and Trends in Information Retrieval (FnTIR), 2022
    (100 page monograph; only student co-author)
5. The Effects of Interactive AI Design on User Behavior: An Eye-tracking Study of Fact-checking COVID-19 Claims
    Li Shi, Nilavra Bhattacharya, **Anubrata Das**, Matt Lease, Jacek Gwizdka
    ACM SIGIR CHIIR, 2022
4. Pareto Solutions vs Dataset Optima: Concepts and Methods for Optimizing Competing Objectives with Constraints in Retrieval
    Soumyajit Gupta, Gurpreet Singh, **Anubrata Das**, Matthew Lease
    ACM SIGIR ICTIR, 2021
3. Fast, accurate, and healthier: Interactive blurring helps moderators reduce exposure to harmful content
    Anubrata Das, Brandon Dang, Matthew Lease
    AAAI HCOMP, 2020
2. Dataset bias: A case study for visual question answering
    Anubrata Das, Samreen Anjum, Danna Gurari

ASIS&T, 2019
(**Diversity and Inclusion Student Best Paper Award** by the School of Information, UT Austin)

1. Predicting trends in the twitter social network: a machine learning approach
   Anubrata Das, Moumita Roy, Soumi Dutta, Saptarshi Ghosh, Asit Kumar Das
   Springer SEMCCO, 2014

Workshops / Book Chapters

7. Proceedings of the 5th Workshop on Trustworthy NLP (TrustNLP 2025)
   Trista Cao, **Anubrata Das**, Tharindu Kumarage, Yixin Wan, Satyapriya Krishna, Ninareh Mehrabi, Jwala Dhamala, Anil Ramakrishna, Aram Galystan, Anoop Kumar, Rahul Gupta, Kai-Wei Chang
   NAACL Workshop Report, 2025
6. Fairness in recommender systems
   Michael D Ekstrand, **Anubrata Das**, Robin Burke, Fernando Diaz
   Recommender systems handbook, Springer, 2022
5. Longhorns at DADC 2022: How many linguists does it take to fool a Question Answering model? A systematic approach to adversarial attacks
   Venelin Kovatchev, Trina Chatterjee, Venkata S Govindarajan, Jifan Chen, Eunsol Choi, Gabriella Chronis, **Anubrata Das**, Katrin Erk, Matthew Lease, Junyi Jessy Li, and others
   Proceedings of the First Workshop on Dynamic Adversarial Data Collection, 2022
4. FACTS-IR: fairness, accountability, confidentiality, transparency, and safety in information retrieval
   Alexandra Olteanu, Jean Garcia-Gathright, Maarten de Rijke, Michael D Ekstrand, Adam Roegiest, ... **Anubrata Das**, ... and others
   SIGIR FACTS-IR Workshop Report, 2021
3. ExFacto: An Explainable Fact-Checking Tool
   Anubrata Das, Sooyong Lee, An Thanh Nguyen, Abhilash Kharosekar, Shankar Krishnan, Saurav Krishnan, Eric Tate, Byron C Wallace, Matthew Lease, and others
   Knight Research Network Tool Demonstration Day, 2021
2. CobWeb: A Research Prototype for Exploring User Bias in Political Fact-Checking
   Anubrata Das, Kunjan Mehta, Matthew Lease
   SIGIR FACTS-IR Workshop, 2019
1. Interactive information crowdsourcing for disaster management using SMS and Twitter: A research prototype
   Anubrata Das, Neeratyoy Mallik, Somprakash Bandyopadhyay, Sipra Das Bit, Jayanta Basak
   PerCom Workshops, 2016

Technical Reports

2. The Case for Claim Difficulty Assessment in Automatic Fact Checking
   Prakhar Singh, **Anubrata Das**, Junyi Jessy Li, Matthew Lease
   arXiv preprint arXiv:2109.09689, 2021
1. A Conceptual Framework for Evaluating Fairness in Search
   Anubrata Das, Matthew Lease
   arXiv preprint arXiv:1907.09328, 2019

Presentations

Invited Talks

Towards Trustworthy AI Through Actionable and Human-Centered Interpretability

- Fundamental AI Research (FAIR) at Meta

- 01/12/2026

Information Assurance in Clinical LLMs through Unlearning

- MISQ Workshop on *Artificial Intelligence-Information Assurance Nexus: The Future of Information Systems Security, Privacy, and Quality*, 07/11/2025

Developing Language Technologies to Complement Human Capabilities

- Microsoft Research FATE Group, New York City, 02/16/2024
- McCombs School of Business, University of Texas at Austin, 02/12/2024

ProtoTEx: Explaining Model Decisions with Prototype Tensors

- Research Colloquium, UT Austin, iSchool, 09/20/2022
- iSchools European Doctoral Seminar Series, 09/16/2022
- Amazon Science Clarify Team, 05/17/2022
- NEC Laboratories Europe, 06/09/2022

Commercial Content Moderation and Psychological Well-Being

- TxHCI - A seminar organized by HCI Researchers across Universities in Texas, 10/02/2020
- Amazon AWS Science, 10/14/2020
- Amazon Human-in-the-loop (HILL) services team, 10/23/2020
- ACM SIGCHI Mumbai Chapter, 26th Meet, 08/28/2021

Conference Presentations

Finding pareto trade-offs in fair and accurate detection of toxic speech. iConference. 2025. Bloomington, Indiana.

ProtoTEx: Explaining Model Decisions with Prototype Tensors. ACL. May 2022. Dublin, Ireland.

You are what you tweet: Profiling users by past tweets to improve hate speech detection. iConference. March 2022. Virtual Conference.

Exfacto: An explainable fact-checking tool. Knight Research Network Tool Demonstration Day, 2021. Virtual Conference.

Fast, Accurate, and Healthier: Interactive Blurring Helps Moderators Reduce Exposure to Harmful Content. AAAI HCOMP 2020. Virtual Conference.

Dataset bias: A case study for visual question answering. ASIS&T 2019. Melbourne, Australia.

CobWeb: A Research Prototype for Exploring User Bias in Political Fact-Checking. ACM SIGIR Workshop on Fairness, Accountability, Confidentiality, Transparency, and Safety in InformationRetrieval (FACTS-IR), 2019. Paris, France.

Local Presentations

ProtoTEx: Explaining Model Decisions with Prototype Layers. Research Colloquium, School of Information, University of Texas at Austin. November 2021. Lightning Talk.

ProtoBART: Explaining Model Decisions with Prototype Layers. TACCSTER: TACC Symposium for Texas Researchers. September 2021. Lightning Talk.

Funding

- Student Professional Development Award for Attending iConference 2025. Award Amount *$750.*
- Evaluating Example-based Explainable Models in Large Language Models. **Amazon AWS Cloud Credit for Research**. Funding period: 11/30/2022 - 11/30/2023. **26,000 USD** (AWS Service Credits).
- Student Professional Development Award for Attending ACL 2022. Award Amount *$1000.*
- **UT Good Systems Grand Challenge —** Graduate Student Grant Proposal. **Anubrata Das**, Chenyan Jia, Shivam Garg. Supervisor: Min Kyung Lee. *Designing algorithmic nudge to reduce inadvertent COVID-19 misinformation sharing on social media*. Awarded - USD 7000.

Service

Workshop Organization

- TrustNLP 2026 at ACL 2026 (Upcoming)
- TrustNLP 2025 at NAACL 2025

Program Committees and Reviewing

- CHI 2026; SIGIR Algorithmic Bias Workshop 2025; CoLM 2024, 2025; ACM FAccT 2025; ACL Rolling Review 2022, 2023, 2024; AAAI AIES 2022; BlackboxNLP Workshop 2022; CHI 2021, 2022; CSCW 2021, 2022, 2023; The Web Conference 2021; Annual Meeting of the Association for Information Science and Technology: 2019, 2020; Information Processing and Management Journal

Conference Volunteer

- NAACL 2025; ACL 2022; CSCW 2019

University Committees

- Assistant Professor Hiring Committee 2020-2021
- Doctoral Studies Committee, School of Information, 2019-2020

Teaching and Mentoring

- Teaching Assistant *Fall 2020*
  - INF385T.3 / CS395T: Human Computation and Crowdsourcing by Dr. Matt Lease
  - Three tutorials on Amazon Sagemaker Ground Truth for collecting data annotations
- Co-Supervising student research with Dr. Matt Lease *01/2022 - 06/2022*
  - Undergraduate thesis on Active Learning with Natural Language Rationales
  - Featured in UT Austin, College of Natural Sciences News
- Co-Supervising undergraduate research group with Dr.Matt Lease *06/2020 - 08/2021*
  - A group of ten students
  - Working on fact-checking using NLP and Human-computation methods

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Research Internships

Cisco Research, Responsible AI — **New York City, NY**
Research Intern — *09/2023 – 12/2023*

- Mentors: Ali Payani, Jayanth Srinivasa

Amazon Nova Responsible AI — **New York City, NY**
Research Intern — *06/2023 – 09/2023*

- Mentors: Kai-Wei Chang, Anna Rumshisky, Aram Galstyan, Manoj Kumar, Ninareh Mehrabi, Anil Ramakrishna, Rahul Gupta

Max Planck Institute of Informatics — **Saarbrücken, Germany**
Research Intern, Databases and Information Systems Group — *06/2019 – 08/2019*

- Mentor: Gerhard Weikum
- Project: *Systematic discovery of bias: A case study on Airbnb Listings*

Indian Institute of Management Calcutta — **Kolkata, India**
Undergraduate Research Assistant, Management Information Systems Group — *10/2012 – 09/2015*

- Mentor: Somprakash Bandyopadhyay

Indian Institute of Technology Kharagpur — **West Bengal, India**
Research Intern, Complex Networks and Research Group — *05/2013 – 07/2013*

- Mentor: Saptarshi Ghosh

Industry Experience

Microsoft — **Hyderabad, India**
Software Engineer — *11/2016 – 07/2018*

- Built and maintained a marketing management tool for Microsoft Universal Store

Mu Sigma — **Bangalore, India**
Decision Scientist — *08/2015 – 10/2016*

- Design and build research prototypes for algorithmic trading using machine learning

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Skills

Research Methodologies: Deep Learning, Large Language Models (LLMs), Foundation Models, Post-training of LLMs, Interpretability, Co-design, Human-AI Interaction
Programming Languages: Python, Javascript
Technologies: Pytorch, Huggingface Transformers, Distributed Training on SLURM, Scikit-Learn, NLTK, SciPy, NumPy, Git
Survey Tools: Qualtrics
Crowdsourcing: Amazon Mechanical Turk, AWS Sagemaker Ground Truth, Prolific
Languages: Fluent in English and Bengali, Knowledge of Hindi